

Trainer Debrief — Lab 25

Response Surface Methodology and Robust Design (Black Belt Only)

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Optimise factor settings using RSM and make the process robust to noise (A5, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-25-rsm-central-composite.xlsx

The 5 centre points reveal significant CURVATURE ($p < 0.001$) — a first-order model is inadequate, which is exactly why you escalate from factorial to RSM. Fitted second-order model $R^2 \sim 0.96$ with significant negative quadratic terms in both factors, so the surface is a dome with an interior MAXIMUM. Stationary point: clamp ~ 47.4 bar, dwell ~ 2.76 s, predicted $Y \sim 242.5$ kPa. Robustness: the surface is flat near the peak, so a ± 1 bar drift costs under 1 kPa — the setting is robust to normal process noise. These are the settings the Lab 26 pilot runs and the Lab 27 control plan holds.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Curvature is tested with centre points, your second-order model is visualised as a contour or surface plot with a classified stationary point, and the robust settings are confirmed under varied noise conditions.

Where this sits in the capstone

BLACK BELT ONLY. Factorial designs find WHICH factors matter; RSM finds the OPTIMUM SETTINGS by modelling curvature that a two-level design cannot see. Taguchi robust design then goes further — setting the process so it performs consistently despite noise you cannot control. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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